

## CO2 - Activity.

1. CCTV image contains salt and pepper noise:

Noise: Salt and pepper.

Technique: Median filter, because it removes impulse noise while preserving edges.

2. Satellite image is corrupted by Gaussian noise

Noise: Gaussian Noise.

Technique: Gaussian filter, because it smooths Gaussian noise effectively.

3. Medical X-ray image has random noise but fine edges must be preserved.

Noise: Random (Gaussian) Noise.

Technique: Bilateral filter, because it reduces noise while preserving edges.

4. Scanned document has black and white dots

Noise: Salt-and pepper noise.

Technique: Median filter, because it removes black and white impulse dots

5. Low-light image shows heavy grain:

Noise: Grain (speckle) Noise.

Technique: Median filter, because it removes black and white impulse dots. Non-local means (NLM) Filter, because it removes grain while preserving details.

6) Foggy road im

Noise: Haze (Fog)

Technique: Histogram visibility.

7) Underwater im

Noise: Scattering

Technique: Histogram image clarity.

8) Chest X-Ray. w

Noise: Low contrast

Technique: Histogram and intensity

9) Night-time ts

Noise: Low light

Technique: Histogram and intensity

10) Grayscale im

Noise: Low contrast

Technique: Histogram values



6) Foggy road image with low contrast.

Noise: Haze (fog noise).

Technique: Histogram equalization - improves contrast and visibility.

7) Underwater image appears dull

Noise: Scattering noise.

Technique: Histogram equalization, enhances contrast and image clarity

8) Chest X-Ray with poor visibility

Noise: Low contrast noise.

Technique: Histogram equalization - increases brightness and intensity variation.

9) Night-time traffic image lacks intensity variation.

Noise: Low illumination noise.

Technique: Histogram equalization - increases brightness and intensity variation.

10) Grayscale image with narrow intensity range

Noise: Low Dynamic Range noise.

Technique: Histogram equalization - spreads intensity values to get better image quality